

Mohd Talib

Senior Full Stack AI Engineer · Tech Lead

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Based in India · Open to remote and relocation (USA · EU · UK · Gulf).

Engineering leader and de facto CTO with 5+ years building and leading cross-functional teams to ship production AI systems end-to-end: **multi-tenant agentic platforms**, **semantic search**, and **MLOps pipelines**. As de facto CTO at **Dumroo.ai**, directed a **20-engineer team** to build Dumroo 2.0, now serving **6 K-12 districts (10+ schools)**, delivering **50% lower inference cost**, **2× accuracy**, **10× faster onboarding**. At **TheAgentic AI**, lead delivery of **10 enterprise platforms for 10 clients in 8 months** and contribute to **CortexON** (455 GitHub stars). Directed a **60+ engineer, 15-country team** at Omdena. Author of the open-source agent framework **OpenFishh**.

EXPERIENCE

Tech Lead · Senior Full Stack AI Engineer Nov 2025 - Present

TheAgentic AI · Applied AI Research, California · Remote

Lead engineer driving architecture and delivery of the company's enterprise AI platforms with a small forward-deployed team, owning client discovery, system design, and the path to production (**2 to 4 weeks each**).

- **CortexON (open source)**. Contributed core multi-agent task planning and execution APIs to TheAgentic's open-source generalized AI agent for everyday task automation: **455 GitHub stars**.
- **Core Platform Engine**. Architected a reusable, multi-tenant agentic orchestration engine (seeding the OpenFishh OSS framework), enabling the rapid deployment of 10 production-grade enterprise platforms for 10 clients within 8 months, while maintaining strict tenant data isolation.
- **Enterprise peer-support platform**. Architected a multi-tenant, privacy-first platform matching support seekers to mentors from voice stories. Built a safety-gated voice-intake pipeline (transcription, moderation + crisis classifier with human-in-the-loop, structured LLM extraction, multi-vector embeddings) feeding a **learned-to-rank matching engine** (pgvector ANN + LightGBM lambdarank that self-retrains from feedback at near-zero inference cost). FastAPI, Postgres/pgvector, async workers, Kubernetes. Enterprise and university pilots in 2026.
- **Reg-tech platform**. Replaced black-box RAG with **deterministic SQL + LLM orchestration**, making compliance workflows auditable and grounding every response in traceable CFR/FDA data.
- **Health-tech platform**. Built an async vision + LLM platform automating metabolic insight generation from medical images through a scalable multi-tenant architecture.
- **Stakeholder Coordination**. Orchestrated weekly design reviews with enterprise client stakeholders, translating complex security and compliance constraints (FDA, CFR) into deterministic graph-based agent workflows.
- **Fundraising-ops platform**. Shipped new features and refactored the codebase of an AI fundraising operating system used by founders across Google for Startups and NVIDIA Inception cohorts.

Senior Full Stack AI Engineer · Tech Lead Mar 2025 - Nov 2025

Dumroo.ai · AI EdTech, New Jersey · Remote · **Promoted to Senior in 6 months** (from Full Stack AI Engineer)

Spearheaded technology strategy as de facto CTO (reporting to CEO), directing a 20-member cross-functional engineering team and managing hiring, developer training, sprint planning, and architectural roadmap execution.

- **Led the team to build Dumroo 2.0 from scratch in 4 weeks**: an AI-driven learning and assessment platform now in production across **US K-12 districts**. Next.js, TypeScript, Supabase, Vercel, GitHub Actions.
- Designed and engineered the core AI infrastructure of Dumroo 2.0, implementing semantic vector caching and a dynamic LLM routing gateway to cut inference costs by 50% while doubling accuracy.
- Partnered with school administrators and district heads to ship role-based access control (RBAC) and automated onboarding pipelines, scaling adoption to **6 districts / 10+ schools in 6 months** with **10+ districts in the pipeline**.
- Architected internal developer productivity and MLOps tooling (automated regression testing and evaluation pipelines), reducing system setup and customer onboarding latency by 10x.
- Integrated Stripe billing with role-based access; deployed CI/CD across Vercel and Netlify via GitHub Actions.

Head of AI & Engineering Apr 2024 - Feb 2025

CodeSpaze · Software, Web & AI Development Services · Remote

Returned to CodeSpaze to lead AI & Engineering after starting here as a Software Engineer in 2021.

- Accelerated the AI Services division to drive a 3x growth in Monthly Recurring Revenue (MRR) by design-patterning reusable ML templates and full-stack services.
- Spearheaded a 20-member cross-functional engineering team (AI, backend, frontend, design, marketing), establishing CI/CD pipelines, automated testing, and developer onboarding to scale capacity for a pipeline of 10,000+ candidates.
- Engineered and deployed full-stack AI systems and internal automation tools, while conducting technical training sessions and mentoring interns in Python, Data Science, and Web Development.

Software Engineer & Data Analyst Jul 2023 - Apr 2024

Techpile Technology Pvt. Ltd. · Lucknow, India · On-site · **Best Performer: Software Developer (2023)**

- Architected high-throughput .NET Core APIs and optimized SQL database schemas, reducing query latency by 40% and improving dashboard loading latency.
- Directed technical delivery of client web services, introducing reusable micro-frontend systems that secured new project renewals and increased services revenue by completing deliverables ahead of schedule.

Machine Learning Engineer to Data Engineering Lead Mar 2023 - Jul 2023

Omdena · Global AI for Social Good, New York · Remote

- Designed and executed a semi-automated active learning labeling pipeline utilizing confidence-threshold pre-annotation, completing a 100k+ healthcare dataset in 3 weeks (accelerating total project timeline by 30%).
- Orchestrated a distributed team of 60+ ML engineers across 15 countries to train, evaluate, and deploy YOLOv8 + PyTorch computer vision pipelines, delivering the diagnostics system to the client ahead of schedule.

Software Engineer Jul 2021 - Feb 2023

CodeSpaze · Software, Web & AI Development Services · Remote

- Delivered **8+ client applications** across web, AI, and automation engagements, owning frontend and backend features end-to-end on **2 to 3 week** delivery cycles.
- Built reusable component libraries and internal tooling that cut team setup and delivery time by **~30%**, and prototyped the early AI/ML work that seeded the company's AI service line.

SELECTED PROJECTS

OpenFishh [Apache 2.0](#) · [OSS](#) [17 stars](#) · [3 forks](#) openfishh.com github.com/MohdTalib0/OpenFishh

Open-source **agent infrastructure platform** built around the three hardest problems in production agents: **agent memory** (persistent, provenance-tracked), **agent reliability** (verification, confidence scoring, safe fallbacks), and **agent evaluation** (knowing when an agent is actually good). The OSS counterpart to HumanizingLabs, Apache 2.0 licensed for ecosystem adoption.

Python · FastAPI · PostgreSQL · pgvector · OpenAI · Anthropic · LangChain · Multi-Agent Orchestration

HumanizingLabs humanizing-labs.vercel.app

A living digital society of **1,200 AI agents across 31 knowledge domains**. Reads the internet continuously, maintains persistent belief memory with full provenance chains, runs structured debates on contested claims, and produces auditable intelligence reports on demand. Every belief is typed, sourced, confidence-decomposed, and lifecycle-governed.

Python · Multi-Agent AI · OpenAI · Anthropic · PostgreSQL · FastAPI · Redis · RAG · LLM Orchestration

Wrively wrively.com [github](#)

AI writing platform that learns your voice and writes from it. Builds a voice profile once, then drafts topic-to-published-post in under 3 minutes. Includes draft orchestration and one-click publish to multiple channels.

Next.js · TypeScript · OpenAI · Supabase

BlueDrum AI bluedrumai.com [beta](#) [github](#)

Legal evidence organization platform. Collects chats, documents, and financial records into structured case files that lawyers can actually use. AI-assisted classification, entity extraction, and timeline reconstruction.

React · FastAPI · AI/ML · PostgreSQL

TalPred AI [github](#)

Production-grade equity alpha system with automated daily cross-sectional predictions, paper trading, monitoring dashboard, and calibration tracking. Designed with proper backtesting discipline and signal decay analysis.

XGBoost · PostgreSQL · Redis · MLflow · React

SKILLS & TOOLS

AI Systems: Agent Orchestration, Agent Memory, Multi-Agent Systems, AI Reliability, Tool Calling, RAG, LLM Evaluation, Provenance & Observability, Fine-tuning, Prompt Engineering, LangChain, OpenAI / Anthropic / Gemini APIs, Vector DBs (pgvector), Vision-LLM

Full-Stack Engineering: TypeScript, Python, Next.js, React, FastAPI, Node.js, Express, Django, Flask, Tailwind CSS, React Native, .NET

Data & ML: PostgreSQL, pgvector, Supabase, Neon, Redis, MongoDB, MySQL, Firebase, SQLAlchemy, Alembic, MLflow, PyTorch, TensorFlow, scikit-learn, XGBoost, Computer Vision (YOLOv8, CNN)

Infrastructure & DevOps: Docker, AWS, Azure, Vercel, Netlify, Render, GitHub Actions, CI/CD, async architectures, REST API design, observability, runbook documentation

Engineering Leadership: Hiring & interviewing, sprint planning, knowledge transfers, US client coordination, mentoring, code review, technical writing, 0-to-1 product delivery

EDUCATION

B.Tech, Computer Science & Engineering 2019 - 2023

Sagar Institute of Technology & Management, Barabanki, India

SELECTED COURSEWORK & CERTIFICATIONS

- Deep Learning Specialization · DeepLearning.AI (Andrew Ng)
- Machine Learning · Stanford University (Coursera)
- Generative AI with Large Language Models · DeepLearning.AI and AWS
- LangChain for LLM Application Development · DeepLearning.AI
- Prompt Engineering for Developers · DeepLearning.AI
- MLOps Specialization · DeepLearning.AI

RECOGNITION & AWARDS

- 2025 **Fastest promotion to Senior Engineer** · at Dumroo.ai, six months from Full Stack AI Engineer to Senior
- 2025 **Apache 2.0 open-source author** · published OpenFishh (17 stars, 3 forks)
- 2023 **Top 10, Data Science Hackathon** · at Innomatics Research Labs
- 2023 **Best Performer, Software Developer** · at Techpile Technology
- 2023 **Letters of Recommendation** · from CodeClause and The Sparks Foundation